AWS NETWORKING PROJECT

VPC Peering

Private Connectivity Between Two Amazon VPCs

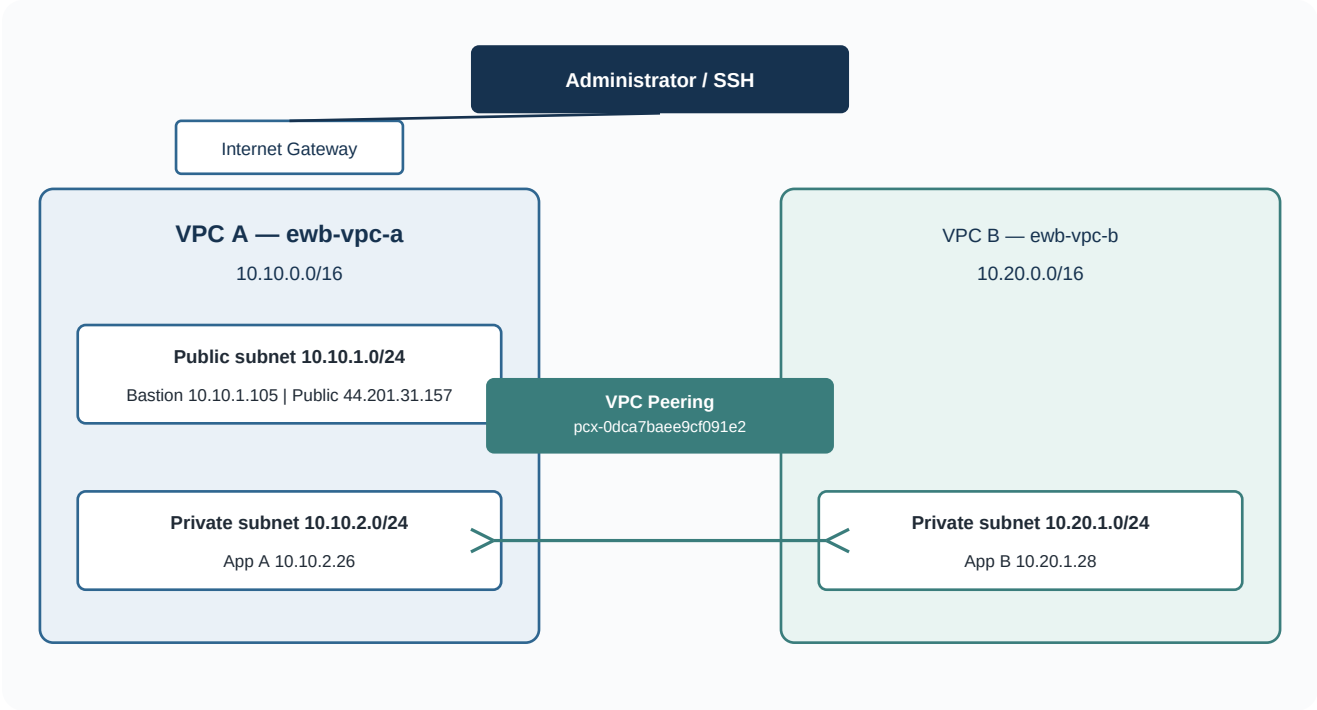

Figure 1. Logical architecture of the implemented VPC peering lab. Diagram is project-specific; console screenshots should be added as evidence where required.

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Batch	[Enter your batch]
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Prepared as a record of the resources configured and connectivity tests performed in the AWS Console and EC2 terminal.

1. Project Overview

The objective of this project was to establish private network communication between two Amazon Virtual Private Clouds (VPCs) using AWS VPC Peering. The design includes a public bastion host in VPC A and two private application instances. The project validates connectivity before and after peering, SSH access through the bastion, and the effect of an ICMP security-group rule.

2. Project Objectives

• Create two VPCs with non-overlapping CIDR ranges.
• Configure public and private subnets and route-table associations.
• Launch a bastion host and two private application instances.
• Establish VPC peering and configure forward and return routes.
• Restrict access with security groups and verify connectivity using ping and SSH.
• Demonstrate controlled ICMP failure and recovery.
• Record evidence and clean up project resources.

3. Network Architecture

VPC A uses CIDR 10.10.0.0/16 and contains a public subnet for the bastion and a private subnet for App A. VPC B uses CIDR 10.20.0.0/16 and contains a private subnet for App B. The VPCs are connected through ewb-a-b-peer. App A and App B communicate using private IPv4 addresses.

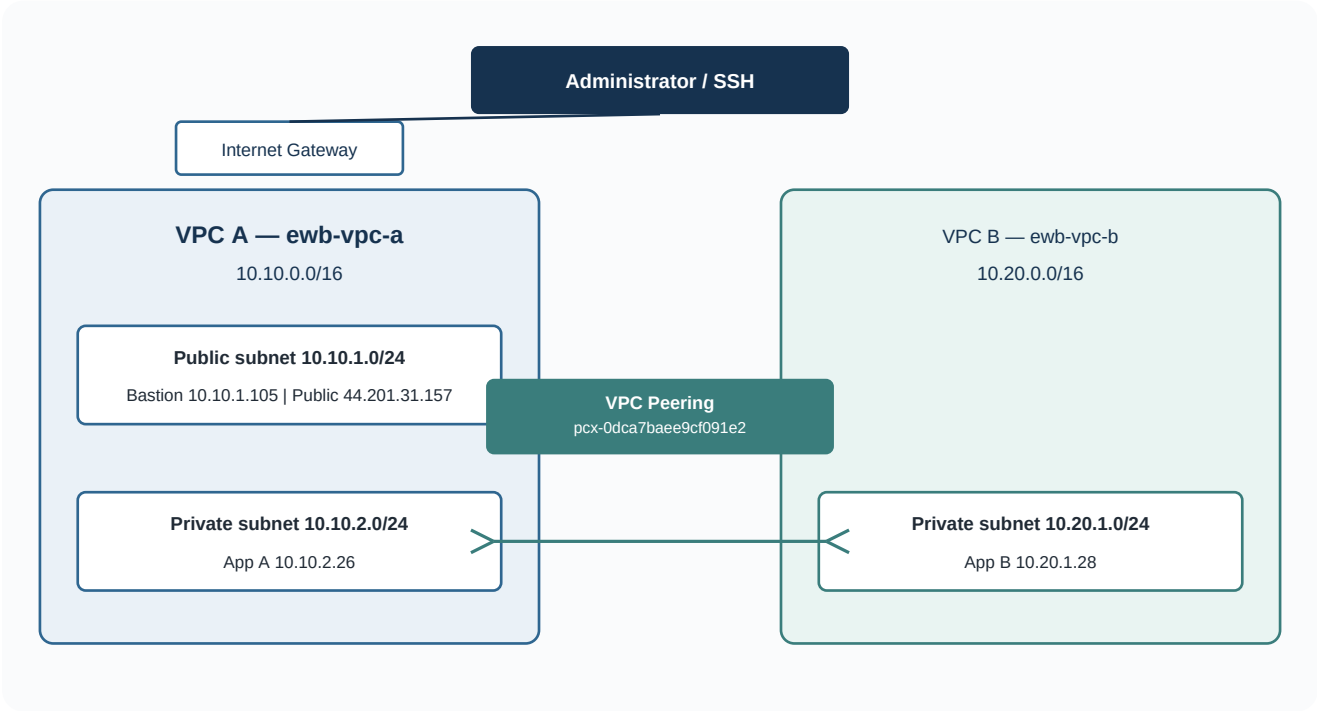

Figure 2. Architecture of the project, including the bastion, private workloads, and peering connection.

4. Resource Inventory

Resource	Name	CIDR / Address / ID
VPC A	ewb-vpc-a	10.10.0.0/16
VPC B	ewb-vpc-b	10.20.0.0/16; ID vpc-0049e7cd6f3c24859
Public subnet A	ewb-a-public	10.10.1.0/24
Private subnet A	ewb-a-private	10.10.2.0/24
Private subnet B	ewb-b-private	10.20.1.0/24; ID subnet-0857473bc0b11f8c0
Internet Gateway	ewb-igw-a	Attached to VPC A

Resource	Name	CIDR / Address / ID
Bastion	ewb-bastion	10.10.1.105; public 44.201.31.157; i-0c6338e47712749d9
App A	ewb-app-a	10.10.2.26; i-00f1c7e7e0324f2bc
App B	ewb-app-b	10.20.1.28; i-069d27a4fa07065f1
Peering	ewb-a-b-peer	pcx-0dca7baee9cf091e2

The two application instances were configured without public IPv4 addresses. The bastion was the administrative entry point.

5. Route-Table Configuration

Three route tables were updated. The local routes were retained. The peering routes provide both the forward path and the return path.

5.1 Public route table — VPC A

Destination	Target	Purpose
10.10.0.0/16	local	VPC A internal traffic
0.0.0.0/0	ewb-igw-a	Internet access for the public subnet
10.20.0.0/16	pcx-0dca7baee9cf091e2	Route from the bastion subnet to VPC B

Route table ID: rtb-0bba294a36df635dc. Associated with ewb-a-public.

5.2 Private route table — VPC A

Destination	Target	Purpose
10.10.0.0/16	local	VPC A internal traffic
10.20.0.0/16	pcx-0dca7baee9cf091e2	Reach VPC B

Route table ID: rtb-00bd1838fc538d95d. Associated with ewb-a-private.

5.3 Private route table — VPC B

Destination	Target	Purpose
10.20.0.0/16	local	VPC B internal traffic
10.10.0.0/16	pcx-0dca7baee9cf091e2	Return traffic to VPC A

Route table ID: rtb-0369543b6193cc68a. Associated with ewb-b-private.

6. Security-Group Configuration

Security group	Inbound rule	Source
ewb-sg-bastion (sg-059c312cd3bdf2e16)	SSH / TCP 22	Administrator public IP /32
ewb-sg-app-a (sg-07ee76a4c2c8dfb73)	SSH / TCP 22	10.10.1.105/32
ewb-sg-app-a (sg-07ee76a4c2c8dfb73)	ICMP Echo Request	10.20.1.28/32
ewb-sg-app-b (sg-0cf381735ddcd2c50)	SSH / TCP 22	10.10.1.105/32
ewb-sg-app-b (sg-0cf381735ddcd2c50)	ICMP Echo Request	10.10.2.26/32

The ICMP rule on ewb-sg-app-a was temporarily removed for the controlled failure test and then restored. SSH rules were retained.

7. Implementation Summary

The VPCs, subnets, route tables, Internet Gateway, security groups, and EC2 instances were created in us-east-1. The VPC peering connection was requested and accepted. Peering routes were added to the private route tables and to the public route table in VPC A. SSH agent forwarding was used to access the private hosts through the bastion without copying the private key onto the bastion.

8. Testing and Validation

The following results were observed during the project. The baseline ping was performed before peering and route configuration; subsequent tests were performed after peering became active and routes were configured.

Test 1 — Baseline ping failure

Source: App A (10.10.2.26) | Destination: App B (10.20.1.28)

```
ping -c 4 10.20.1.28
4 packets transmitted, 0 received, 100% packet loss
```

Result: Expected failure before peering was configured.

Test 2 — Forward ping: App A to App B

```
ping -c 4 10.20.1.28
4 packets transmitted, 4 received, 0% packet loss
rtt min/avg/max/mdev = 0.154/0.268/0.586/0.183 ms
```

Result: Passed. Average round-trip time was 0.268 ms.

Test 3 — Reverse ping: App B to App A

```
ping -c 4 10.10.2.26
4 packets transmitted, 4 received, 0% packet loss
rtt min/avg/max/mdev = 0.159/0.182/0.210/0.019 ms
```

Result: Passed. Average round-trip time was 0.182 ms.

Test 4 — SSH to App B through the bastion

```
ssh ec2-user@10.20.1.28
Amazon Linux 2023 login banner displayed
Shell prompt: [ec2-user@ip-10-20-1-28 ~]$
```

Result: Passed. App B was accessed through the bastion host using its private IP address.

Test 5 — Controlled ICMP failure

```
ping -c 4 10.10.2.26
4 packets transmitted, 0 received, 100% packet loss
```

The inbound ICMP Echo Request rule on ewb-sg-app-a was temporarily removed. Ping from App B to App A then failed as expected.

Test 6 — Connectivity recovery

```
ping -c 4 10.10.2.26
4 packets transmitted, 4 received, 0% packet loss
rtt min/avg/max/mdev = 0.170/0.185/0.199/0.011 ms
```

The ICMP rule was restored and ping connectivity returned. Average round-trip time was 0.185 ms.

Test summary

Test	Observed result	Status
Baseline ping	100% packet loss	Expected failure
App A → App B	0% packet loss	Passed
App B → App A	0% packet loss	Passed
SSH through bastion to App B	Successful shell login	Passed
ICMP rule removed	100% packet loss	Expected failure
ICMP rule restored	0% packet loss	Passed

9. Technical Explanation

VPC peering: Provides private IPv4 routing between two VPCs. Peering does not automatically add routes to route tables.

Return routes: Responses need a valid route back to the originating VPC. Both sides must have the appropriate routes.

Security groups: Routes establish a path, but security groups independently permit or block inbound traffic. The ICMP experiment demonstrated this distinction.

Bastion host: Provides an administrative entry point to private instances. App A and App B use private addresses for inter-VPC traffic.

Peering limitation: VPC peering is non-transitive. A peering connection also does not provide VPC B internet access through VPC A's Internet Gateway.

10. Challenges and Solutions

Challenge	Solution
Private instances were not directly reachable from the administrator's computer	Used the bastion host for SSH access.
Ping initially failed	Created the peering connection and configured routes.
Private key was not available on the bastion	Used SSH agent forwarding; the private key was not copied to the bastion.
ICMP connectivity failed during the test	Restored the inbound ICMP Echo Request rule.
Linux ping syntax failed in Windows PowerShell	Ran ping -c 4 inside the Linux EC2 SSH session.

11. Lessons Learned

The project provided hands-on experience with CIDR planning, public and private subnets, route-table associations, VPC peering, return routes, security-group source restrictions, bastion-host access, SSH agent forwarding, ICMP testing, and systematic troubleshooting. It also showed that successful connectivity depends on routes, peering state, security rules, and instance configuration working together.

12. Cleanup and Cost Management

Status: Pending verification. Do not mark cleanup complete until the resources have actually been removed and checked in the AWS Console.

Cleanup checklist:

1. Terminate ewb-bastion, ewb-app-a, and ewb-app-b.
2. Check and remove leftover project EBS volumes or snapshots.
3. Release any Elastic IP created for the lab and delete the lab key pair if no longer needed.
4. Remove the VPC peering connection and peering routes as required.
5. Delete the lab security groups after their network interfaces disappear.
6. Delete the project subnets and custom route tables.
7. Detach and delete ewb-igw-a.
8. Delete ewb-vpc-a and ewb-vpc-b.
9. Review the Region and AWS Billing for remaining project resources or charges.

13. Conclusion

The project demonstrated private communication between two VPCs in the us-east-1 Region. Following peering activation and route configuration, App A and App B successfully exchanged ICMP traffic using their private IP addresses. SSH access to App B through the bastion was also successful. The controlled ICMP test showed that removing the inbound rule blocked ping traffic and restoring the rule recovered connectivity. The project reinforced the importance of route symmetry, least-privilege security rules, testing, evidence collection, and responsible resource

cleanup.

14. Screenshot and Evidence Appendix

Insert the following actual AWS Console and terminal screenshots if available. The architecture diagram in this report is a custom illustration; it is not an AWS Console screenshot.

Figure 3 — VPC inventory showing ewb-vpc-a and ewb-vpc-b with their CIDR ranges.

Figure 4 — Subnets showing 10.10.1.0/24, 10.10.2.0/24, and 10.20.1.0/24.

Figure 5 — VPC peering connection ewb-a-b-peer (pcx-0dca7baee9cf091e2) showing Active status.

Figure 6 — The three route tables and their final routes.

Figure 7 — EC2 instance inventory and private IP addresses.

Figure 8 — Baseline ping failure, forward ping success, reverse ping success, SSH login to App B, ICMP failure, and connectivity recovery.

Figure 9 — Cleanup evidence showing the project resources removed. Add only after cleanup is complete.

Security note: Never include private keys, passwords, access tokens, or other credentials in the report.

Report prepared from the project configuration and test outputs recorded by the student. Verify any resource status that may have changed before submission.